

Nehal Garg

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EDUCATION

Masters of Science in Computer Science

May 2026

University of Southern California | GPA: 3.56

Courses: Algorithm Analysis, Web Technologies, Applied Natural Language Processing, Machine Learning for Data Science, Foundations of Modern Data Management and Processing, Software Engineering, Affective Computing

Bachelor of Technology in Computer Science

June 2024

Guru Gobind Singh Indraprastha University | GPA: 3.72

Courses: Data Structures, Object Oriented Programming, Operating Systems, Artificial Intelligence, Distributed and Parallel Systems

EXPERIENCE

Software Engineer Intern

June 2025 - Sept 2025

ZofAI (Next.js, TypeScript, Tailwind CSS, Prisma, PostgreSQL, Vercel, Python Integration)

San Francisco, CA

- Built full-stack features (dashboards, access control, usage analytics) that enabled 5+ client teams to automate QA workflows.
- Designed PostgreSQL schemas and Prisma models to improve data reliability and cut query times by 30% during load testing.
- Deployed systems on Google Cloud (GCP) using Docker and CI/CD workflows on Unix-based environments.

Software Engineer Intern

Sept 2023 - Nov 2023

Essell 24 (TypeScript, React.js, REST APIs, Git, CI/CD, Postman)

San Francisco, CA

- Optimized core asset pipeline using lazy loading and compression in React.js, reducing page load latency by 16%.
- Collaborated with backend engineers to redesign REST API responses, reducing redundant payloads and improving responsiveness.
- Built and unit-tested scalable UI components in TypeScript for CI/CD pipelines, improving reusability and deployment reliability.

Software Developer Intern

June 2023 - Aug 2023

Hammer Lifestyle (React Native, Redux, Node.js, Firebase, REST APIs, MongoDB)

Haryana, India

- Developed a health-tracking module for smartwatches using React Native and Firebase, reducing sync failures by 24%.
- Optimized data sync logic in Firebase backend, improving real-time reliability across 10K+ active users.
- Collaborated with backend engineers to optimize MongoDB queries, enhancing data ingestion pipeline performance and reducing processing time.

Software Developer Intern

March 2023 - May 2023

Nyadeko (React.js, TypeScript, Node.js, MongoDB, JWT, V8 Optimization)

New Delhi, India

- Redesigned performance-critical dashboard using React.js and TypeScript, increasing workflow efficiency across B2B users.
- Developed and secured RESTful endpoints with token-based multi-layer authentication and persistent session handling.
- Reduced API response latency by 35% through logic refactoring and async handling in dashboard workflows, boosting efficiency.

PROJECTS

Hospital Management System (React.js, GoLang, PostgreSQL, REST APIs, JWT)

[\[GitHub\]](#)

- Built a fullstack hospital system to streamline patient workflows with role-based access and appointment scheduling.
- Developed GoLang REST APIs with JWT authentication to enable scalable multirole operations and secure medical data handling.
- Designed normalized PostgreSQL schemas to optimize appointment and record management, reducing redundant queries by 25%.
- Implemented responsive React dashboards for realtime hospital insights, improving operational visibility and workflow efficiency.

Artist Discovery Android App (Kotlin, Jetpack Compose, Retrofit, JWT)

[\[GitHub\]](#)

- Created a modern Android app to search, view, and favorite artists using the Artsy API with JWT-based secure authentication.
- Persisted user interactions (favorites) using MongoDB and managed UI state using Jetpack Compose and ViewModel architecture.
- Optimized network responsiveness through coroutine-based Retrofit calls and async UI handling.

EmoTunes – AI Music Recommendation Engine (Python, TensorFlow, OpenCV, Spotify API)

[\[GitHub\]](#)

- Engineered a real-time emotion recognition system using CNN trained on FER-2013, achieving 85% accuracy across 7 emotions.
- Integrated with Spotify API to generate mood-based playlists, reducing manual search and increasing engagement.
- Combined OpenCV-based face tracking with async audio logic and real-time feedback to ensure responsive music transitions.

RESEARCH PUBLICATION

AppleGo – Product Discovery Platform (React.js, Django, MySQL, Google Maps API, REST APIs)

[\[Publication Link\]](#)

- Analyzed 50K+ product queries to evaluate retailer visibility and proposed a scalable backend for real-time inventory comparison.

Blockchain-based Crowdfunding Framework (IJISRT) (Solidity, Node.js, PostgreSQL, REST APIs, Git)

[\[Publication Link\]](#)

- Proposed a blockchain crowdfunding framework using Solidity smart contracts to improve transparency and transaction security.

TECHNOLOGIES

Programming Languages: Java, Python, C++, JavaScript, TypeScript, Kotlin, C, XML, GoLang

Frameworks / Libraries: React, Redux, Node.js, Express, Django, Flask, Servlet, Bootstrap, TensorFlow, PyTorch, OpenCV

Tools/DevOps/Databases: Git, Docker, Postman, AWS, GCP, HTML, CSS, MATLAB, MongoDB, MySQL, PostgreSQL, Unix/Linux, SQL, NoSQL

Concepts: REST APIs, JWT Auth, Distributed Systems, Agile, Version Control, SDLC

Machine Learning: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn